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2026 CHALLENGE

Post-Quantum Secure Communication Architecture for Industrial Robotic Production Cells

Operator and Technician Documentation

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This document provides operational and maintenance guidance for the StratoTech Post-Quantum Secure Communication Architecture for Industrial Robotic Production Cells. It is intended for production floor operators, maintenance technicians, and integration engineers. No cryptography background is required.

Glossary

Term	Definition
OT	Operational Technology. The physical hardware layer — sensors, actuators, and microcontrollers on the production floor.
GA	Global Aggregator. The gateway laptop that verifies all incoming sensor data before forwarding it to the dashboard.
IT	Information Technology. The monitoring and data processing layer — the dashboard laptop.
ESP32	The Wi-Fi enabled microcontroller used as the OT sensor node in this prototype.
Telemetry	Live sensor data sent from the OT device to the gateway and dashboard.
ML-DSA	A quantum-resistant algorithm that creates a digital signature on every sensor message. Ensures the data has not been changed in transit.
Private Key	A secret code stored on each OT device used to sign outgoing sensor messages. Must never be shared or copied.
Public Key	The non-secret part of the key pair included in the device certificate. Used by the gateway to verify signatures.
Certificate	A digital identity document issued to each OT device by the Certificate Authority. Proves the device is authorised to send data.
CA	Certificate Authority. Issues and validates certificates for all OT devices in the system.
PKI	Public Key Infrastructure. The system that manages device certificates and keys to confirm device identity.
PQC	Post-Quantum Cryptography. Security algorithms designed to remain secure even against quantum computers.
NIST	National Institute of Standards and Technology. The US body that standardised the quantum-safe algorithms used in this system in August 2024.
Freshness Window	The maximum age of a packet the gateway will accept. In this prototype it is set to 5 seconds. Older packets are rejected as potentially unsafe.
Verification Runtime	The time the gateway takes to verify each incoming packet. Measured in milliseconds.
Memory Overhead	The amount of gateway laptop memory used by the security layer.
TEE	Trusted Execution Environment. A secure area inside a processor that protects sensitive data such as private keys from being extracted even under physical attack.

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Term	Definition
ARM TrustZone	A TEE technology built into ARM processors that can protect private keys from physical extraction.
IEC 62443	International standard for cybersecurity in industrial automation and control systems.
CISA	Cybersecurity and Infrastructure Security Agency. US government body that publishes cybersecurity guidelines for industrial environments.

Table 1: Glossary of Terms

Contents

1 Introduction

This document provides operational and maintenance guidance for the StratoTech Post-Quantum Secure Communication Architecture for Industrial Robotic Production Cells. The system protects sensor data communications across three layers — the Operational Technology layer, the Global Aggregator, and the Information Technology layer, using quantum-resistant security algorithms standardised by NIST in August 2024.

It is intended for three audiences: production floor operators who monitor the Streamlit dashboard, maintenance technicians who configure and maintain OT devices, and integration engineers responsible for deploying the system into existing factory infrastructure.

Each section is labelled with its intended audience. No cryptography background is required. A glossary of all technical terms is provided at the end.

2 System Overview

The system uses physical hardware components connected across three layers. At the sensor level, a small microcontroller reads distance measurements and controls a motor representing a robotic actuator. This data is sent to a gateway laptop which checks that the data is genuine and has not been changed. Only verified data is passed to a monitoring laptop where operators can see the system status on a live dashboard.

The table below lists each physical component, what it is, and what it does in the system.

Component	What It Is	What It Does
ESP32	Wi-Fi enabled microcontroller	Collects sensor readings and sends them securely to the gateway
Ultrasonic Sensor (HC-SR04)	Distance measuring sensor	Measures distance and generates live readings from the production cell
Servo Motor (SG90)	Small PWM-controlled motor	Drives the robotic actuator and simulates machine movement
Linear Actuator	3D printed rack-and-pinion mechanism	Controls straight-line motion in the production cell
Gateway Laptop	Runs `aggregator.py`	Receives sensor data, verifies it is genuine, and forwards only trusted messages to the dashboard
Monitoring Laptop	Runs `dashboard.py`	Displays live sensor readings, security status, and alerts to the operator

Table 2: Hardware Components and Architecture Roles

3 Dashboard Operation Guide

Intended audience: Production floor operators. No cryptography knowledge required.

The dashboard allows operators to monitor accepted packets, blocked attacks, and system status without needing to understand the cryptographic details. If an alert appears, the operator should follow the response guide below and contact a technician if needed.

3.1 Status Indicators

Indicator	Meaning	Action Required
SECURE	Security checks are valid and telemetry is trusted.	No action required.
ACTIVE	The OT device, GA, and dashboard are running.	Continue monitoring.
Accepted Packets	Number of packets accepted by the GA.	Check that the value increases during operation.
Blocked Attacks	Number of invalid or malicious packets blocked.	Inform a technician if it increases repeatedly.
Verification Runtime	Time taken by the GA to verify packets.	Report unusually high values.
Memory Overhead	Memory used by security functions.	Report unstable or high usage.

Table 3: Dashboard Status Indicators

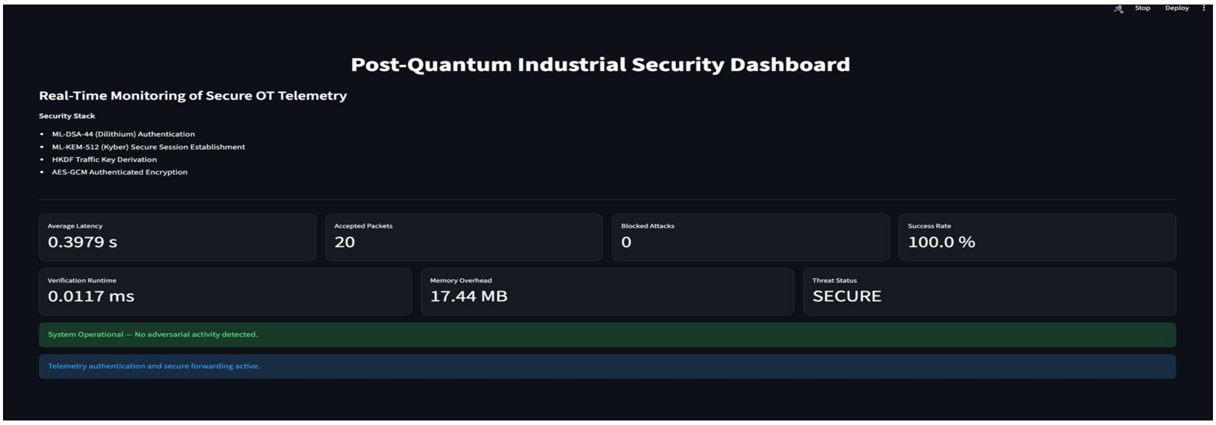

Figure 1: Live Dashboard — Status Indicators View

3.2 Alert Response Guide

WARNING

If any alert appears, follow the procedures below. Do not modify system configuration without contacting a qualified technician.

Alert	Meaning	Action
REPLAY ATTACK BLOCKED	An old packet was resent using a repeated counter.	Record the alert and inform a technician.
INVALID ML-DSA SIGNATURE	The packet data or signature is not valid.	Stop trusting that telemetry stream and report it.
FAKE CERTIFICATE	The device certificate is invalid or forged.	Disconnect the suspected device and contact support.
STALE PACKET	The packet arrived too late.	Check network delay if the alert repeats.
UNAUTHORIZED DEVICE	An unregistered device tried to send telemetry.	Reject the device and verify the enrolment list.

Table 4: Alert Response Guide

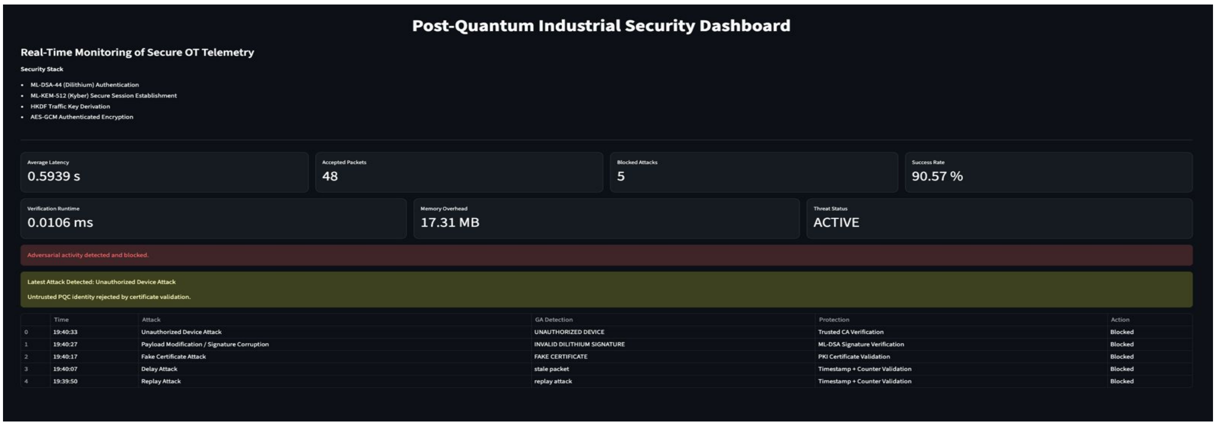

Figure 2: Live Dashboard — Alert Notifications View

4 Configuration Guide

Intended audience: Maintenance technicians responsible for configuring and maintaining OT devices.

4.1 Key Management

WARNING

Private keys must never be transmitted, copied, or shared. If a key is suspected to be compromised, take the device offline immediately and contact the security team.

- **Protect private keys:** The private key must stay only on the authorised OT device. It must not be copied, shared, or moved to another device.
- **Use valid certificates:** Each OT device must have a certificate signed by the trusted Certificate Authority. The gateway will reject unknown, expired, or modified certificates.
- **Revoke compromised devices:** If a device is compromised, replaced, or removed, remove its certificate from the trusted devices list in `aggregator.py` immediately.
- **Limit access:** Only authorised technicians should modify keys, certificates, or the trusted devices list in `aggregator.py`. All changes should be logged.

4.2 Timestamp Configuration

- **Synchronise clocks:** The OT device, gateway laptop, and monitoring laptop should use synchronised clocks. If clocks are out of sync, valid packets may be incorrectly rejected.
- **Freshness window:** The system rejects packets older than 5 seconds. This time limit is called the freshness window. Do not change this value without testing in a non-production environment first.
- **Check timestamp errors:** If many packets are being rejected as stale, check the ESP32 clock, gateway clock, and network delay before changing any settings.

4.3 Device Enrolment Procedure

1. **Assign device ID:** Give the OT device a unique identifier such as `OT_DEVICE_01`.
2. **Generate keys:** Run the key generation script on the ESP32 to create a new ML-DSA key pair. The private key stays on the device. The public key is included in the device certificate.
3. **Register device at gateway:** Add the device certificate to the trusted devices list in `aggregator.py` on the gateway laptop.
4. **Test enrolment:** Send one valid telemetry packet and confirm the gateway accepts it and the dashboard shows it as received.

5 Maintenance Guide

5.1 Routine Checks

Check	Frequency	Action
Dashboard status	Every shift	Check that telemetry, packet count, and security status are updating normally
Blocked attack count	Daily	Review blocked packet logs and investigate any repeated errors
Certificate expiry	Monthly	Check certificate validity in `aggregator.py` and renew any certificates expiring within 30 days
Network connectivity	Weekly	Verify connection between the ESP32, gateway laptop, and monitoring laptop

Table 5: Routine Maintenance Checks

5.2 Troubleshooting

Problem	Likely Cause	Solution
All packets rejected	Wrong certificate, wrong public key, or trusted devices list mismatch	Check certificate and trusted devices list in `aggregator.py`. Re-enrol the device if needed
Dashboard not updating	`dashboard.py` stopped or gateway is not forwarding data	Restart `dashboard.py` and check the gateway logs
No packets received	ESP32 offline, sensor issue, or wrong gateway IP or port in `ot_signer.py`	Check ESP32 power, Wi-Fi, and confirm the IP address and port in `ot_signer.py` match the gateway settings
Authentication failure	Modified payload, invalid signature, fake certificate, or unknown device	Verify the device certificate and confirm the packet is signed by the correct private key
High latency alerts	Network delay, high packet rate, or gateway overload	Reduce packet rate, check Wi-Fi signal strength, and monitor gateway performance

Table 6: Troubleshooting Guide

6 Safety Precautions

6.1 Physical Security

WARNING

Any suspected physical tampering must be reported immediately. Disconnect the device from the network and do not reconnect until a security check is complete.

Physical access to the ESP32 allows extraction of the private key, compromising the entire authentication system. One practical mitigation is integrating a Trusted Execution Environment such as ARM TrustZone into the prototype to protect private keys from extraction under physical attack.

- Do not leave the ESP32, breadboard, or connected hardware unattended in an unsecured area
- Do not allow anyone outside the authorised team to handle the hardware
- If the device appears to have been moved, disconnected, or interfered with, take it offline and notify the responsible person before reconnecting

In a real industrial deployment, OT devices would require locked enclosures, tamper-evident seals, and hardware-backed key storage as required by IEC 62443-4-2.

6.2 Operational Security

WARNING

Do not make any configuration changes while the system is running. Stop the process first, make the change, verify it, then restart.

Incorrect configuration can silently break authentication without any visible error on the dashboard.

- Never share private keys or device certificates with anyone outside the authorised team
- Do not change timestamp settings or key assignments while the system is running
- Write down what you are changing and why before making any configuration change

In a real industrial deployment, all configuration changes would require formal approval, logging, and version control as required by CISA Cybersecurity Performance Goals 2.0 and IEC 62443-4-2.

6.3 Network Security

WARNING

Do not connect the ESP32 or gateway laptop to a public or shared network. Use a dedicated network for system operation at all times.

Connecting to an unsecured network exposes the sensor devices to remote attack without requiring physical access.

- Run the system on a dedicated local network. Do not connect the ESP32 or gateway laptop to a public or shared Wi-Fi network during operation
- Only the person responsible for system maintenance should have access to the gateway laptop running `aggregator.py`

- If you see any unexpected alert or connection on the dashboard, report it immediately and do not attempt to resolve it alone

In a real industrial deployment, the OT network would require full segmentation from general IT traffic using network zones and conduits as defined by IEC 62443-3-2.